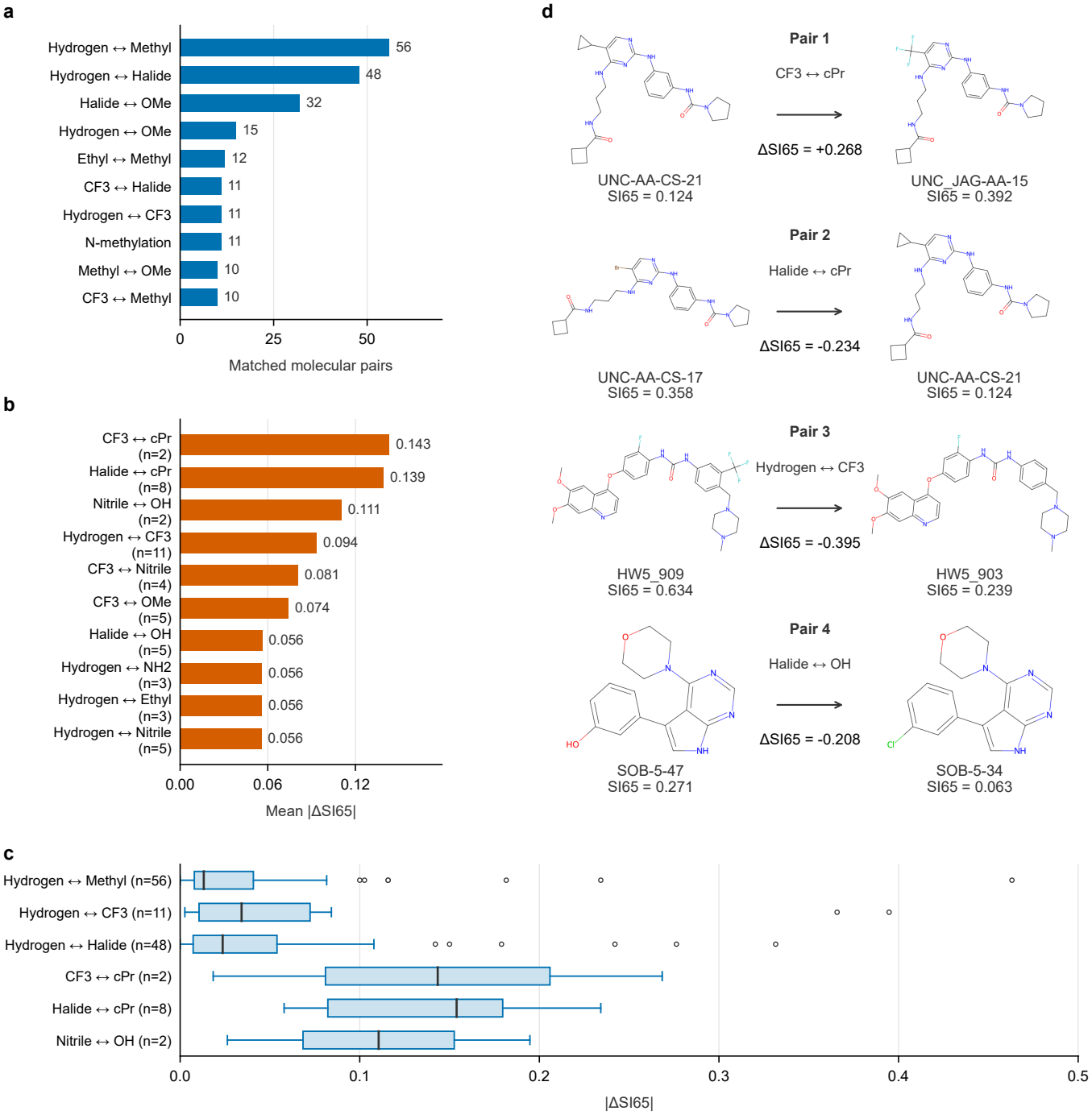

a
Hydrogen ↔ Methyl
56
Hydrogen ↔ Halide
48
Halide ↔ OMe
32
Hydrogen ↔ OMe
15
Ethyl ↔ Methyl
12
CF3 ↔ Halide
11
Hydrogen ↔ CF3
11
N-methylation
11
Methyl ↔ OMe
10
CF3 ↔ Methyl
10
0
25
50
Matched molecular pairs
b
CF3 ↔ cPr (n=2)
0.143
Halide ↔ cPr (n=8)
0.139
Nitrile ↔ OH (n=2)
0.111
Hydrogen ↔ CF3 (n=11)
0.094
CF3 ↔ Nitrile (n=4)
0.081
CF3 ↔ OMe (n=5)
0.074
Halide ↔ OH (n=5)
0.056
Hydrogen ↔ NH2 (n=3)
0.056
Hydrogen ↔ Ethyl (n=3)
0.056
Hydrogen ↔ Nitrile (n=5)
0.056
0.00
0.06
0.12
Mean |ΔSI65|
c
Hydrogen ↔ Methyl (n=56)
Hydrogen ↔ CF3 (n=11)
Hydrogen ↔ Halide (n=48)
CF3 ↔ cPr (n=2)
Halide ↔ cPr (n=8)
Nitrile ↔ OH (n=2)
0.0
0.1
0.2
0.3
0.4
0.5
|ΔSI65|
d
Pair 1
CF3 ↔ cPr
ΔSI65 = +0.268
UNC-AA-CS-21
SI65 = 0.124
UNC_JAG-AA-15
SI65 = 0.392
Pair 2
Halide ↔ cPr
ΔSI65 = -0.234
UNC-AA-CS-17
SI65 = 0.358
UNC-AA-CS-21
SI65 = 0.124
Pair 3
Hydrogen ↔ CF3
ΔSI65 = -0.395
HW5_909
SI65 = 0.634
HW5_903
SI65 = 0.239
Pair 4
Halide ↔ OH
ΔSI65 = -0.208
SOB-5-47
SI65 = 0.271
SOB-5-34
SI65 = 0.063